Assignment Questions

MAT361

1. Addition Rule of Probability

Consider two events A and B such that $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$.

(a) Find $P(A \cup B)$ using the addition rule of probability.

(b) Are the events A and B mutually exclusive? Justify your answer.

2. Conditional Probability

A factory produces electronic devices, 10% of which are defective. A quality control process is 90% effective in detecting defective items and 95% effective in correctly identifying non-defective items. Suppose an item is tested, and the test indicates that it is defective. What is the probability that the item is indeed defective? Use conditional probability.

3. Bayesian Probability

A certain disease occurs in 1% of the population. A test for the disease has a 99% sensitivity (the probability of testing positive given that one has the disease) and a 95% specificity (the probability of testing negative given that one does not have the disease).

(a) If a person tests positive, what is the probability that they actually have the disease? Use Bayes' Theorem to compute this.

(b) Interpret the result in terms of false positives.

4. Discrete Random Variable

Consider a discrete random variable X representing the number of heads obtained when flipping a fair coin three times.

(a) Write the probability mass function (PMF) of X.

(b) Compute $P(X = 2)$.

(c) Find the expected value of X.

5. Continuous Random Variable

Let X be a continuous random variable with probability density function (PDF)

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Verify that $f(x)$ is a valid PDF.

(b) Compute $P(0.25 \leq X \leq 0.75)$.

(c) Find the expected value of X.